

PHD 2025 Batch 2

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Deep Learning

Due date for Submission 22.06.2026

Sr No	Question	Marks
1	Explain the role and significance of activation functions in neural networks. Compare ReLU, Sigmoid, and Tanh with respect to vanishing gradients and computational efficiency.	1
2	Why is proper weight initialization important in training deep networks? Explain Xavier and He initialization strategies.	1
3	Describe any two commonly used loss functions in classification tasks and justify when to use each	1
4	Define overfitting and list at least three techniques to prevent it in neural networks.	1

5	Differentiate between Batch, Stochastic, and Mini-Batch Gradient Descent with examples.	1
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